CSE 676 – Assignment 1

Vision Transformer Assignment Report

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Introduction:

In this report I post the results of a Vision Transformer trained on balanced CIFAR-10 subset. Through ablation studies on patch size, network depth and positional encoding the model was evaluated. Furthermore, attention entropy, positional embedding correlation, per-class accuracy, top confusion pairs and mean attention distance were calculated for quantitative analysis. The baseline model was able to reach a final validation accuracy of 0.4507 with 142,026 trainable parameters.

Section A: Ablation Analysis:

A.1 Patch Size

The model was evaluated with patch sizes of 2, 4, and 8. The final validation accuracies were:

- Patch size 2: 0.4582
- Patch size 4: 0.4507
- Patch size 8: 0.3854

The best result came from using a patch size of 2, and the worst result came from using a patch size of 8. This shows that smaller patches help the model get more detailed information about the area around it. The transformer can see more detailed spatial structure when the patch size is smaller because the image is broken up into more tokens. But this also makes it more expensive to run because self-attention has to work with more tokens.

Larger patches cut down on the number of tokens, which makes the math easier, but they also take away useful fine-grained detail. This is why the model with a patch size of 8 was much less accurate. In general, the results show a clear trade-off: smaller patches make things easier to represent, while larger patches make things more efficient but hurt performance.

A.2 Number of Layers

The model was evaluated with 2, 4, and 8 transformer layers. The final validation accuracies were:

- 2 layers: 0.4420
- 4 layers: 0.4507
- 8 layers: 0.4409

The model which has four layers worked the best. Adding more layers from 2 to 4 made the model a little more accurate, which suggests that a small increase in depth helps the model learn more complex representations. But making the depth even deeper, to 8 layers, didn't make things better; in fact, it made them a little worse.

This shows that accuracy does not always get better as depth increases on this small dataset. Some possible reasons are overfitting, harder optimization in deeper transformers, and not having enough training data. With this problem setup, 4 layers seems to be the best balance between how much the model can learn and how easy it is to train.

A.3 Positional Encoding

The model without positional encoding achieved a final validation accuracy of:

- No positional encoding: **0.4334**

By removing positional encoding made things worse compared to the baseline accuracy of 0.4507. This is to be expected because a Vision Transformer doesn't care about the order of the input data. The model can't tell where a patch came from in the image if the order of the patches isn't encoded.

Self-attention can connect tokens with each other, but it still needs positional information to understand how things are arranged in space. The decrease in accuracy shows that positional encoding is important for classifying images because it gives model information about how patches are arranged and how they relate to each other in space.

Section B: Quantitative Analysis Interpretation :

B.1 Attention Entropy

The per-layer attention entropy values were:

- Layer 0: 5.7618
- Layer 1: 5.5467
- Layer 2: 5.9340

- Layer 3: 6.0035

CLS token attention entropy quantifies the concentration (or dispersion) of the CLS attention distribution. More focused attention means lower entropy; more diffuse attention means higher entropy. In these results, entropy decreases slightly from layer 0 to layer 1, then increases in deeper layers.

This indicates that the model first becomes a bit more selective and then shifts towards more global attention with increasing depth. That is, earlier layers seem to encode more localized information while later layers aggregate information more broadly across the image. This agrees with the hypothesis that the deeper transformer layers learn more global contextual representations.

B.2 Positional Embedding Correlation

The positional embedding correlation result was:

- Pearson correlation: -0.3122
- Number of pairs: 2016

The negative Pearson correlation implies that spatially closer patches tend to have more similar learned positional embeddings. This is a desired property as it shows that the model has learnt some notion of spatial locality. The magnitude is moderate rather than super strong , which means the learned positional embeddings capture spatial structure to a meaningful extent, but not perfectly.

The minus sign is, in general, the most important part of the result. This implies that the positional representation of the model is consistent with spatial proximity, i.e., the nearby patches are embedded more similarly than the distant ones.

B.3 Per-Class accuracy and top-3 Confusions:

The per-class accuracies are:

- Class 0: 0.489
- Class 1: 0.457
- Class 2: 0.228
- Class 3: 0.328
- Class 4: 0.423
- Class 5: 0.380
- Class 6: 0.439

- Class 7: 0.550
- Class 8: 0.687
- Class 9: 0.526

The highest-accuracy classes are classes 8,7 and 9 and the lowest-accuracy classes are 2,3 and 5.

Below are the top-3 confusion pairs:

- [3, 5, 275]
- [0, 8, 243]
- [1, 9, 231]

The results show that some classes are much easier for the model to learn than others. The model is doing great on class 8 and pretty good on classes 7 and 9, but is having a very hard time on class 2, class 3. The high confusion pairs illustrate that the model has challenge in separating some visually similar classes. Perhaps because of the shared textures, shapes or general look-and-feel of those categories.

Overall per-class results indicate that the model does not perform uniformly over the classes of CIFAR-10. Some classes are easier as they have more distinct visual features, while other classes are harder as their representations overlap more in the learned feature space.

B.4 Mean Attention Distance:

The per-layer mean attention distances were:

- Layer 0: 4.0420
- Layer 1: 3.9448
- Layer 2: 4.1610
- Layer 3: 4.2422

Mean attention distance is the average distance each query patch attends over the spatial grid. A low value implies more local attention, a high value implies more global attention. In these results, the value slightly decreases from layer 0 to layer 1 and then increases through layers 2 and 3.

This pattern indicates that the model attends more locally in the beginning and it expands its receptive field in deeper layers. In the last layer, attention is given to more distant patches, suggesting that the model is taking into account more context from the

image. This is similar to convolutional neural networks, where later layers tend to encode more global information relative to earlier layers

Therefore, the effective receptive field tends to increase as the depth grows, which is consistent with the interpretation from the entropy results that deeper transformer layers build more global representations.

Conclusion:

The baseline Vision Transformer earned a final validation accuracy of 0.4507 with 142,026 parameters. The best accuracy (0.4582) was obtained with the patch size 2 in the ablation study, confirming the importance of smaller patches for the preservation of fine-grained spatial information. Adding layers improved performance from 2 to 4 layers but did not help at 8 layers, indicating diminishing returns on this small dataset. Removing positional encoding leads to performance drop, confirming the importance of spatial information for ViTs.

The results from the quantitative side mostly back up what we saw earlier. I think the attention entropy and that mean attention distance point to deeper layers pulling in info from farther away, more globally sort of.It is interesting how the negative correlation in positional embeddings suggests the model picked up on spatial stuff that actually means something position wise.Some classes just seem way easier based on the per class accuracy, and the confusion matrix highlights how visually similar ones get mixed up a lot. That part stands out because it shows not everything is equal.

Overall the Vision Transformer looks like it is working as it should, grabbing local details along with bigger picture stuff. Across the ablations and this quantitative look, the behavior comes off as meaningful, though some bits might need more digging. The global shift in deeper layers repeats that idea of information gathering spreading out.